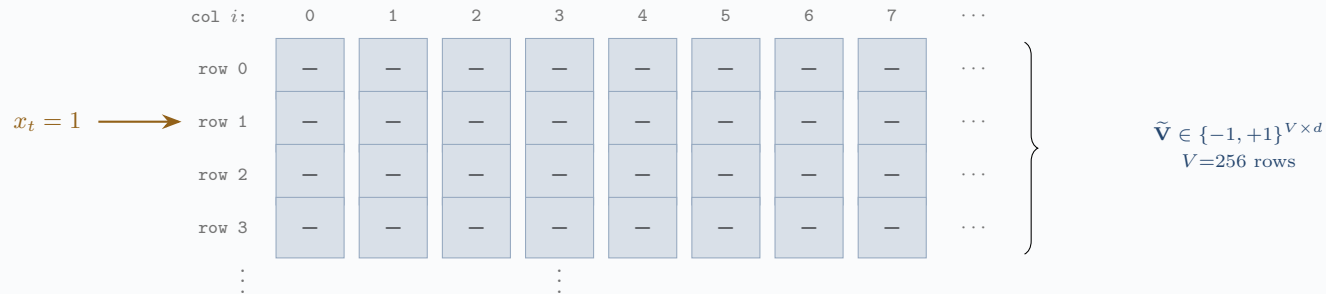
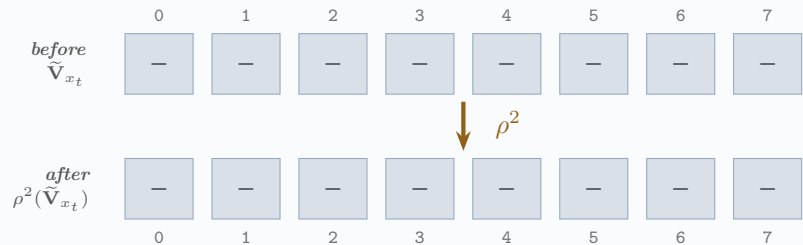


Block 1. Token lookup — bipolar embedding table



Block 2. Position binding ρ^t (cyclic right-rotation)



Cell at index j takes value from index $(j - t) \bmod d$. Different positions produce orthogonal versions of the same token.

$$\rho^t(\mathbf{h})_j = \mathbf{h}_{(j-t) \bmod d} \quad (\text{parameter-free, preserves } \{-1, +1\}^d)$$